



Simplified RC4 Stream Cipher

Let us consider the stream cipher RC4, but instead of the full 256 bytes, we will use 8×3 -bits. That is, the state vector S is 8×3 -bits. We will operate on 3-bits of plaintext at a time since S can take the values 0 to 7, which can be represented as 3 bits.

Assume we use a 4×3 -bit key of $K = [1\ 2\ 3\ 6]$. And a plaintext $P = [1\ 2\ 2\ 2]$

- Initialize the state vector S and temporary vector T
- Perform the initial permutation on S .

```
j = 0;
for i = 0 to 7 do
  j = (j + S[i] + T[i]) mod 8
  Swap(S[i], S[j]);
end
```

- generate 3-bits at a time, k , that we XOR with each 3-bits of plaintext to produce the ciphertext. The 3-bits k is generated by:

```
i, j = 0;
while (true) {
  i = (i + 1) mod 8;
  j = (j + S[i]) mod 8;
  Swap (S[i], S[j]);
  t = (S[i] + S[j]) mod 8;
  k = S[t]; }
```

- Calculate the Ciphertext C corresponding to the plaintext P

Assume we use a 4 x 3-bit key of $K = [1\ 2\ 3\ 6]$. And a plaintext $P = [1\ 2\ 2\ 2]$

- Initialize the state vector S and temporary vector T
- Perform the initial permutation on S .

```

j = 0;
for i = 0 to 7 do
j = (j + S[i] + T[i]) mod 8
Swap(S[i], S[j]);
end

```

a)

step	T	1 ¹	2 ²	3	6	1	2	3	6	i	j	
0	S =	0	1	2	3	4	5	6	7			
1	S =	1	0	2	3	4	5	6	7	0	1	$j = 0 + 0 + 1 = 1$
2	S =	1	3	2	0	4	5	6	7	1	3	$j = 1 + 0 + 2 = 3$
3	S =	2	3	1	0	4	5	6	7	2	3	$j = 3 + 2 + 3 = 8 \times 8 = 0$
4	S =	2	3	1	6	4	5	0	7	3	0	$j = 0 + 0 + 6 = 6$
5	S =	2	3	1	4	6	5	0	7	4	6	$j = 6 + 4 + 1 = 11 \times 8 = 3$
6	S =	2	3	5	4	6	1	0	7	5	3	$j = 3 + 5 + 2 = 10 \times 8 = 2$
7	S =	2	3	5	4	6	0	1	7	6	2	$j = 2 + 0 + 3 = 5$
8	S =	2	3	7	4	6	0	1	5	7	5	$j = 5 + 7 + 6 = 18 \times 8 = 2$

After initial Permutation

$S = [2 \ 3 \ 7 \ 4 \ 6 \ 0 \ 1 \ 5]$

c. generate 3-bits at a time, k , that we XOR with each 3-bits of plaintext to produce the ciphertext. The 3-bits k is generated by:

```
i, j = 0;
while (true) {
    i = (i + 1) mod 8;
    j = (j + S[i]) mod 8;
    Swap (S[i], S[j]);
    t = (S[i] + S[j]) mod 8;
    k = S[t]; }
```

i	j	S = [0 1 2 3 4 5 6 7]	t	K[i]
0	0	2 3 7 4 6 0 1 5	-	-
1	3	2 4 7 3 6 0 1 5	7	$S[7] = 5$
2	2	2 4 7 3 6 0 1 5	6	$S[6] = 1$
3	5	2 4 7 0 6 3 1 5	3	$S[3] = 0$
4	3	2 4 7 6 0 3 1 5	6	$S[6] = 1$

d. Calculate the Ciphertext C corresponding to the plaintext P

$$\begin{array}{r}
 P = [1 \ 2 \ 2 \ 2] = \begin{array}{cccc} 001 & 010 & 010 & 010 \end{array} \\
 K = [5 \ 1 \ 0 \ 1] = \begin{array}{cccc} 101 & 001 & 000 & 001 \end{array} \quad (+) \\
 \hline
 C = \begin{array}{cccc} 100 & 011 & 010 & 011 \end{array} \\
 C = [4 \quad 3 \quad 2 \quad 3]
 \end{array}$$